Task 5: Exploratory Data Analysis (EDA)

Data Analyst Internship — Findings Report

Objective: Explore a Titanic-style passenger dataset using statistical summaries and visualizations to identify patterns, trends, missing values and possible outliers.

Dataset limitation: The CSV supplied in this project is simulated data created for demonstration. It is not the original Titanic/Kaggle dataset. Findings below describe only this simulated sample and should not be represented as historical Titanic findings.

1. Dataset overview

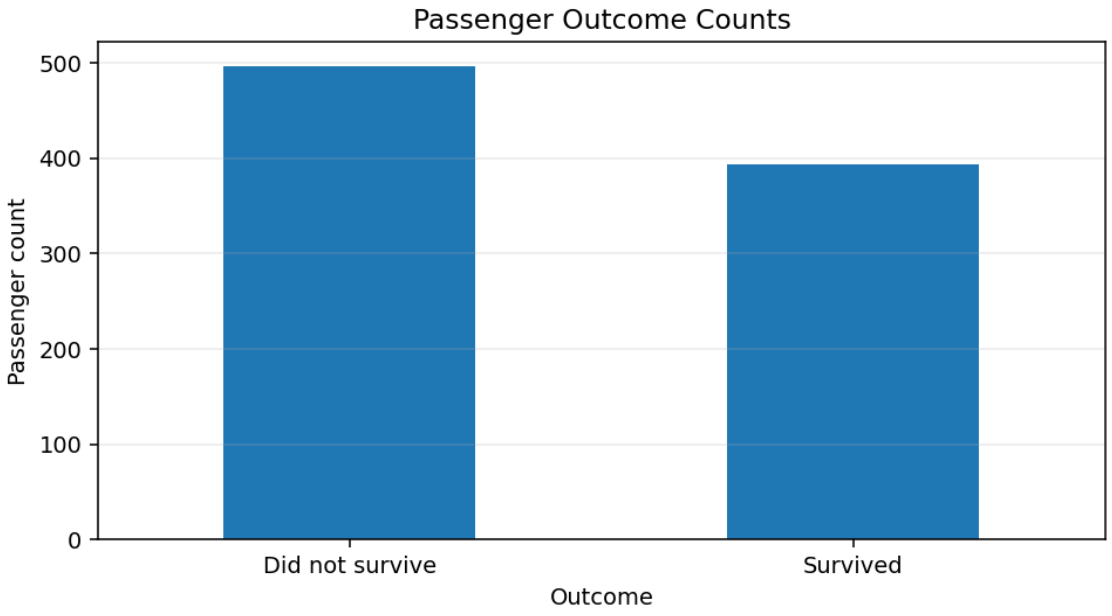
The dataset contains 891 rows and 9 columns. Columns include survival outcome, passenger class, sex, age, family counts, fare and embarkation point. `describe()`, `info()` and `value_counts()` are included in the notebook for reproducible inspection.

2. Data quality

Column	Missing values
Age	178
Embarked	2

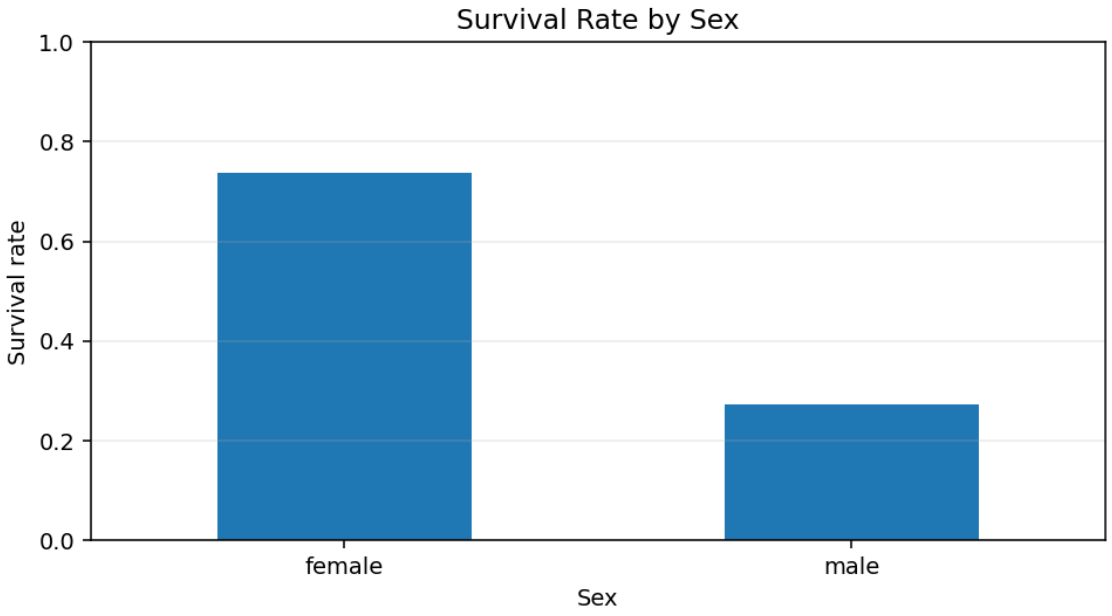
Age has approximately 20.0% missing values in this simulated dataset. Missing data should be assessed before analysis; age could be imputed using a suitable group median or retained as missing depending on the analytical goal. The notebook also checks duplicate rows.

3. Outcome distribution

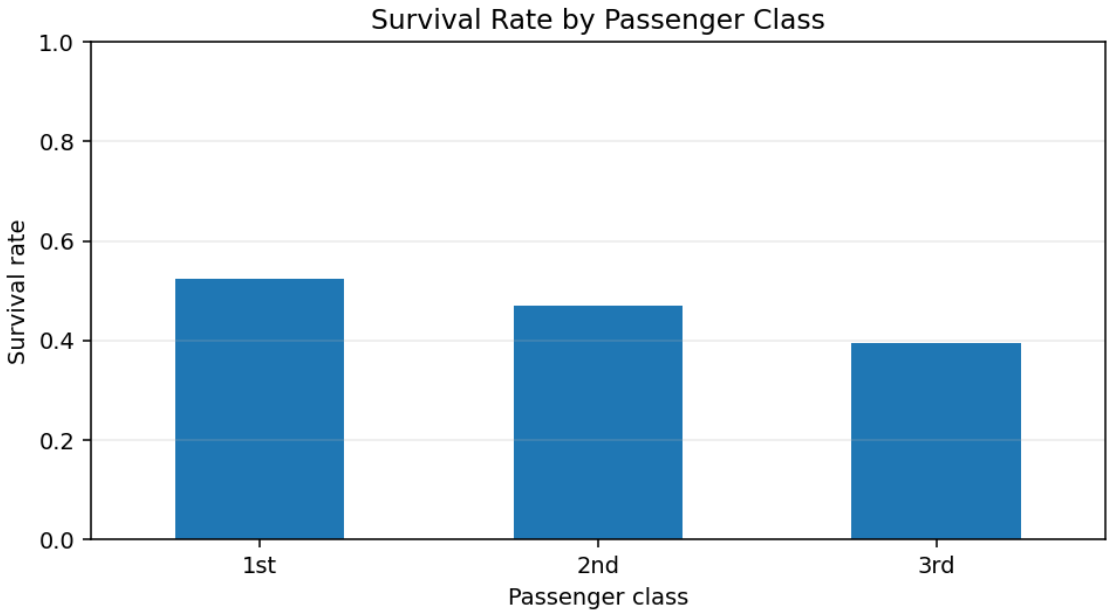


The overall survival rate in this simulated sample is 44.2%. This is a property of the generated data, not a historical Titanic estimate.

4. Survival by sex and class

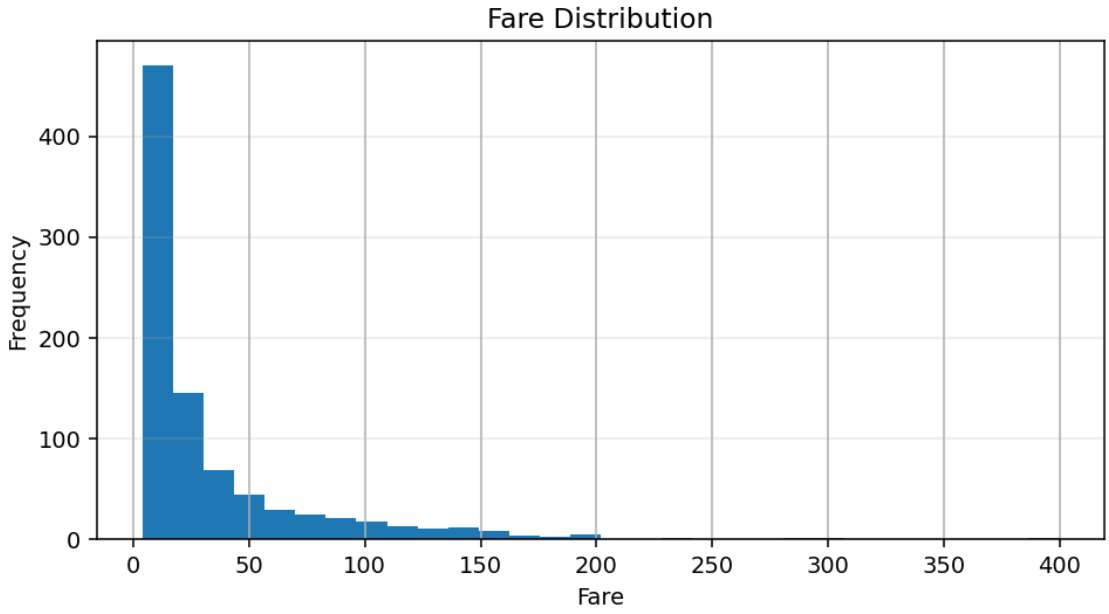


The survival-rate-by-sex chart compares group averages. Because the dataset was simulated with survival probability depending partly on sex, a difference between groups is expected. This is an example of how bivariate analysis can reveal a relationship.

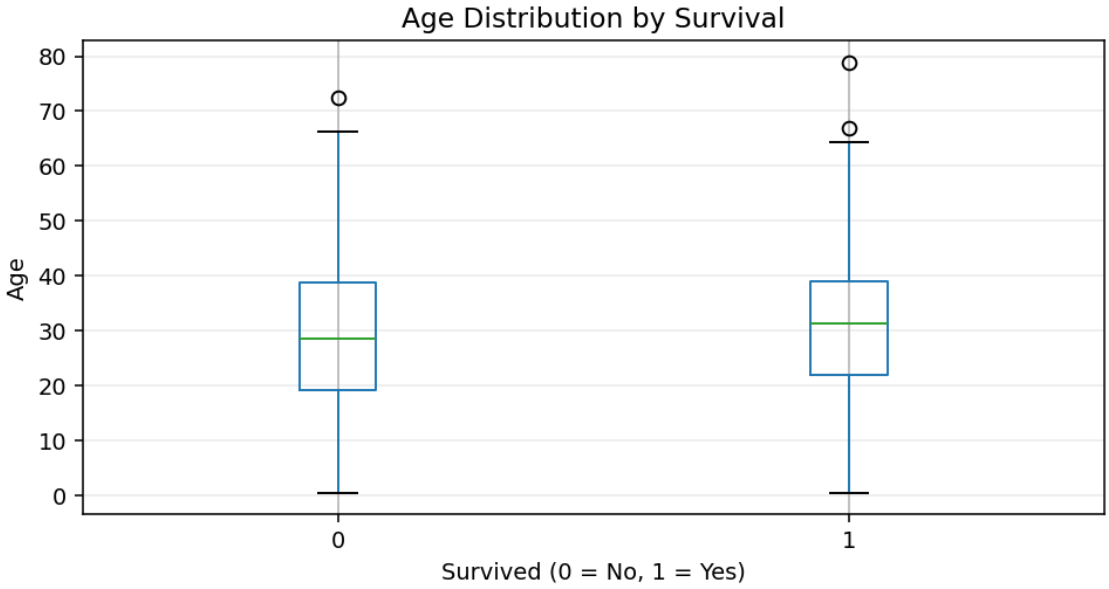


The passenger-class chart compares survival rates across first, second and third class. Differences are descriptive associations in this synthetic dataset; they do not establish causation.

5. Distributions and outliers

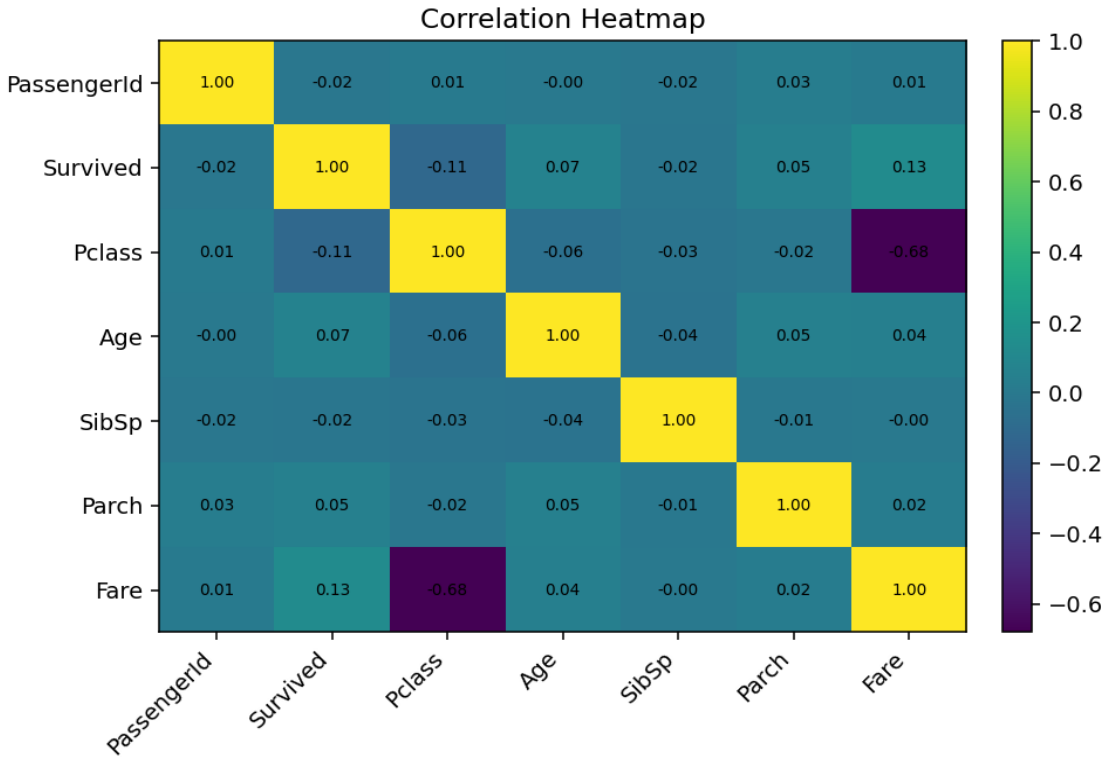


Fare skewness is 3.22. A positive value indicates right-skew, where a smaller number of higher fares extend the distribution to the right. Histograms help inspect distribution shape; a boxplot helps identify spread and potential outliers.



The age boxplot compares median and spread across the two survival groups. It should be interpreted alongside missing age values and group sizes.

6. Correlation heatmap



A correlation heatmap gives a compact view of pairwise linear relationships among numeric columns. Correlation values range from -1 to +1. Correlation does not imply causation, and numeric codes such as passenger class should be interpreted carefully.

7. Conclusions and next steps

• Use .info(), .describe() and .value_counts() to understand structure and category counts.

• Check missingness and duplicates before drawing conclusions; document any cleaning decisions.

• Use histograms and boxplots for single-variable distributions and potential outliers.

• Use bar charts, scatterplots, pairplots and heatmaps to explore relationships.

• For a production or historically accurate report, replace the simulated data with the official Titanic dataset and rerun the notebook.

8. Interview questions

What is EDA and why is it important? EDA uses statistics and visualizations to understand data quality, patterns and anomalies before modeling.

Which plots check correlation? Scatterplots and correlation heatmaps; pairplots help inspect multiple numeric relationships.

How do you handle skewed data? Inspect the shape and outliers; consider transformations such as log for suitable positive variables and explain the choice.

How do you detect multicollinearity? Check predictor correlations and calculate Variance Inflation Factor (VIF) when appropriate.

Univariate, bivariate and multivariate analysis? Analysis of one variable, two variables, and three or more variables.

Difference between heatmap and pairplot? A heatmap summarizes a matrix of values; a pairplot displays pairwise plots and individual distributions.

How do you summarize insights? State the finding, support it with values/plots, explain limitations and relate it to the business question.